

Linkit Kim/SMD

linkit-v4

Low-power wildlife telemetry platform designed for long-term field deployment.
This variant integrates the SMD Argos module, optimized power management (wireless charging, solar input, and USB/Li-ion handling), and a compact architecture prioritizing energy efficiency, cost, and reliability. All interfaces and components are arranged to ensure seamless compatibility with the Linkit firmware ecosystem and Arribada's open-source tooling.

Table of contents

Sheet 1: Project overview (this page)

Sheet 2: linkit-v4

Sheet 3: pwr

Sheet 4: BQ51003YFPT_WirelessCharger_2.5W (devicesheet)
Sheet 5: TPS63901_BuckBoost_lowIq_3V32V3_400mA (devicesheet)

Sheet 6: mcu

Sheet 7: sensors
Sheet 8: BMA400_ACC_LinkitV3 (devicesheet)
Sheet 9: SAM-M10Q (devicesheet)

Sheet 10: interface

Sheet 11: satellite
Sheet 12: sat-smd

Legend

General comment such as function title, configuration, ...

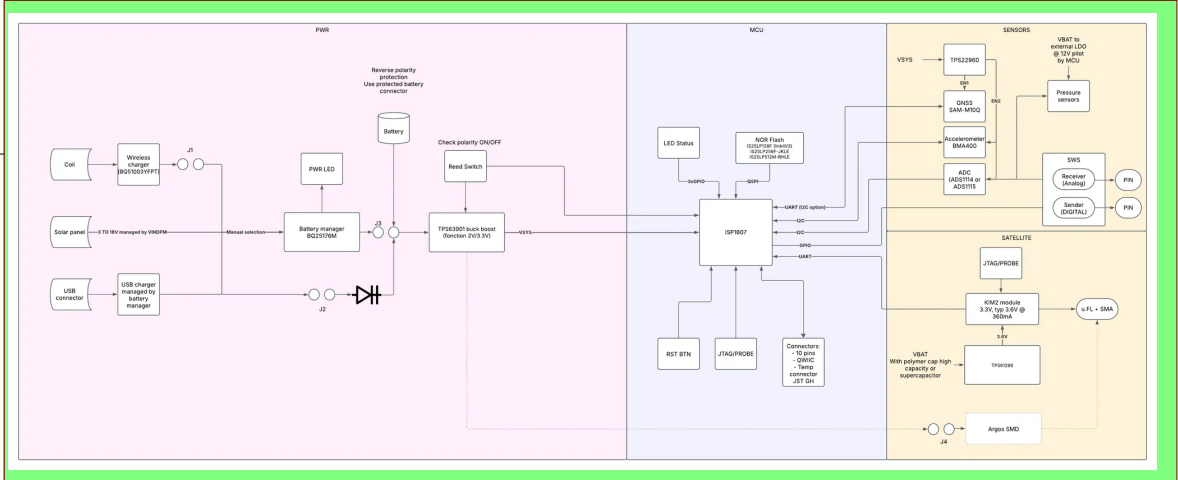
Text to be added to silkscreen.

Warning text.

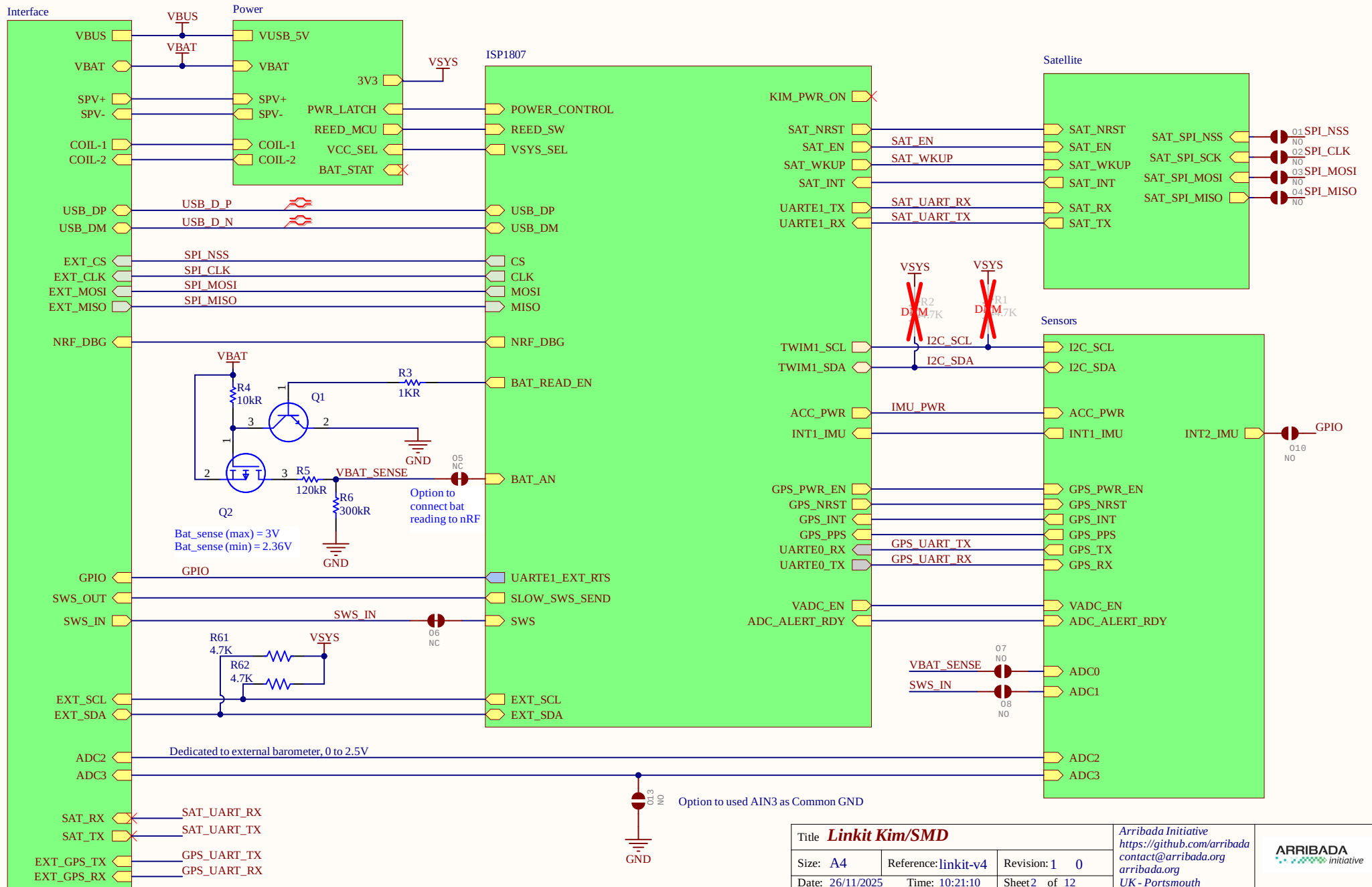
Notes to generate the board layout.

License

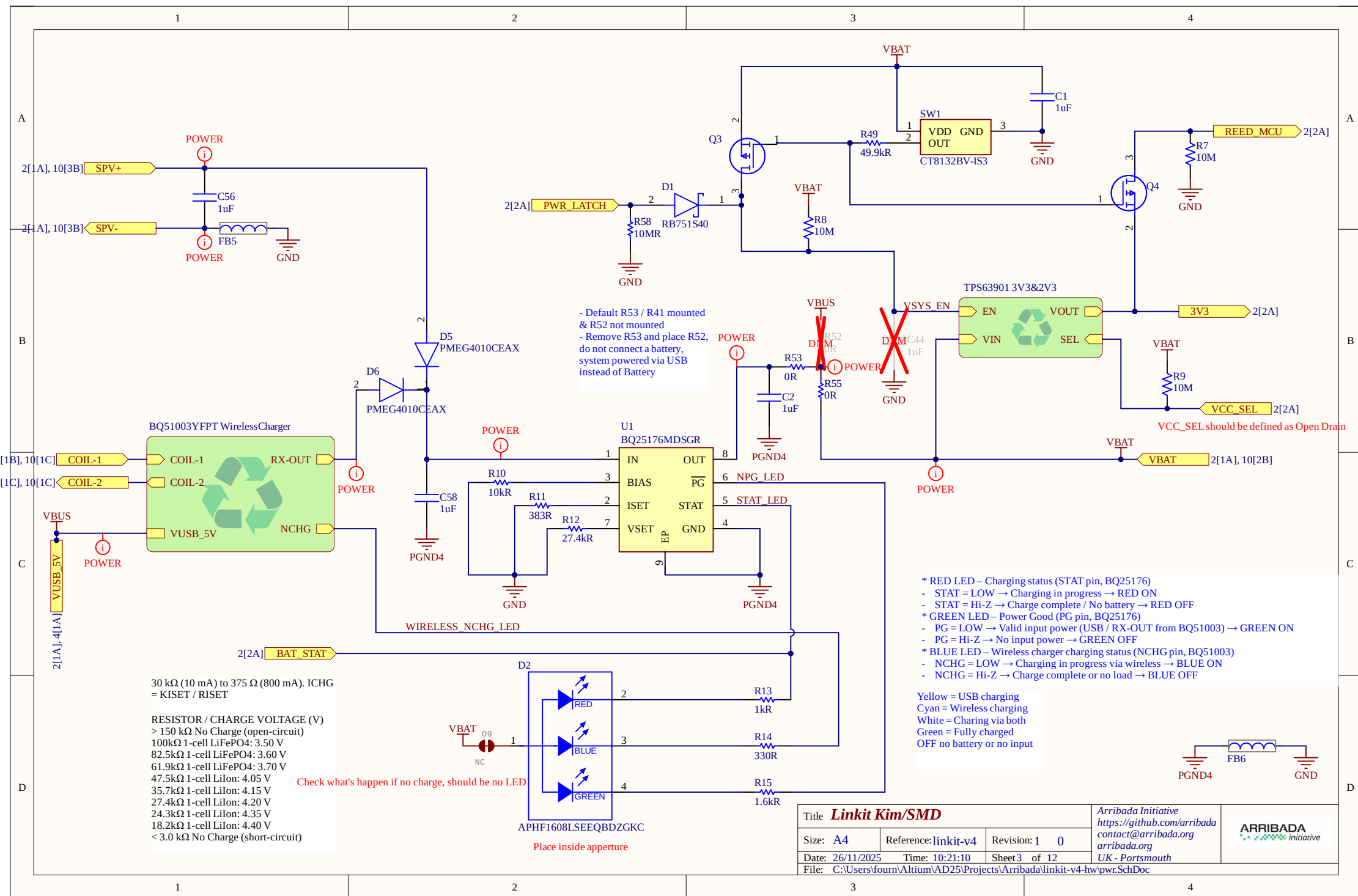
U_linkit-v4
linkit-v4.SchDoc

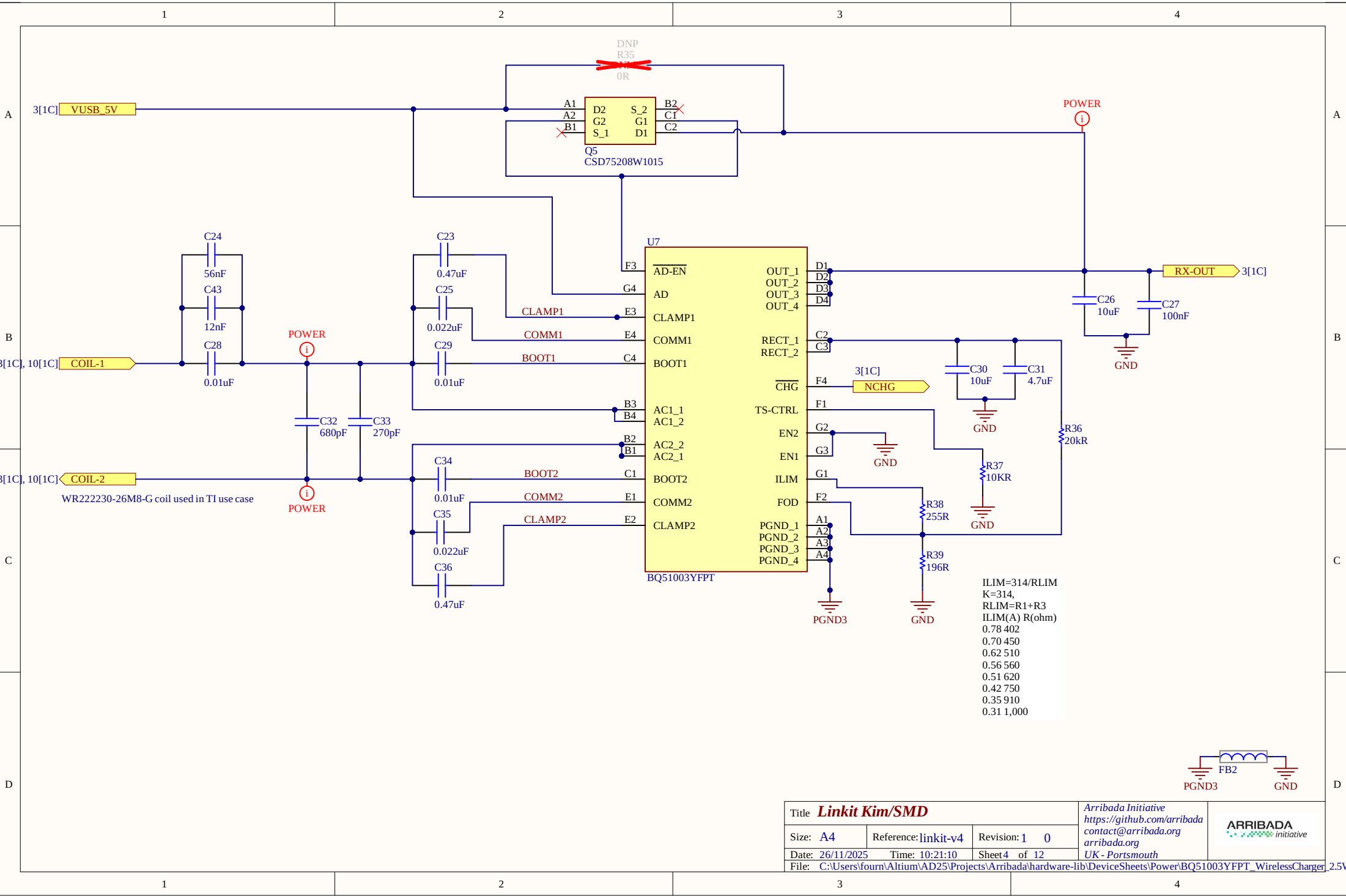


Title Linkit Kim/SMD			Arribada Initiative https://github.com/arribada contact@arribada.org arribada.org UK - Portsmouth		
Size: A4	Reference: linkit-v4	Revision: 1 0			
Date: 26/11/2025	Time: 10:21:09	Sheet 1 of 12			
File: C:\Users\fourm\Altium\AD25\Projects\Arribada\linkit-v4-hw\Project_Overview.SchDoc					



Title Linkit Kim/SMD			Arribada Initiative https://github.com/arribada contact@arribada.org arribada.org		
Size: A4	Reference: linkit-v4	Revision: 1 0			
Date: 26/11/2025	Time: 10:21:10	Sheet 2 of 12	UK - Portsmouth		
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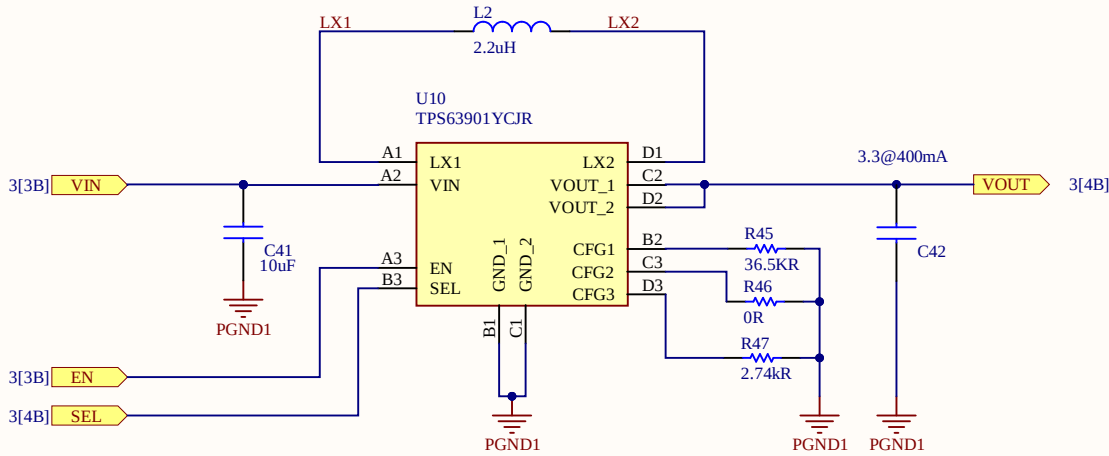
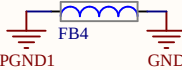
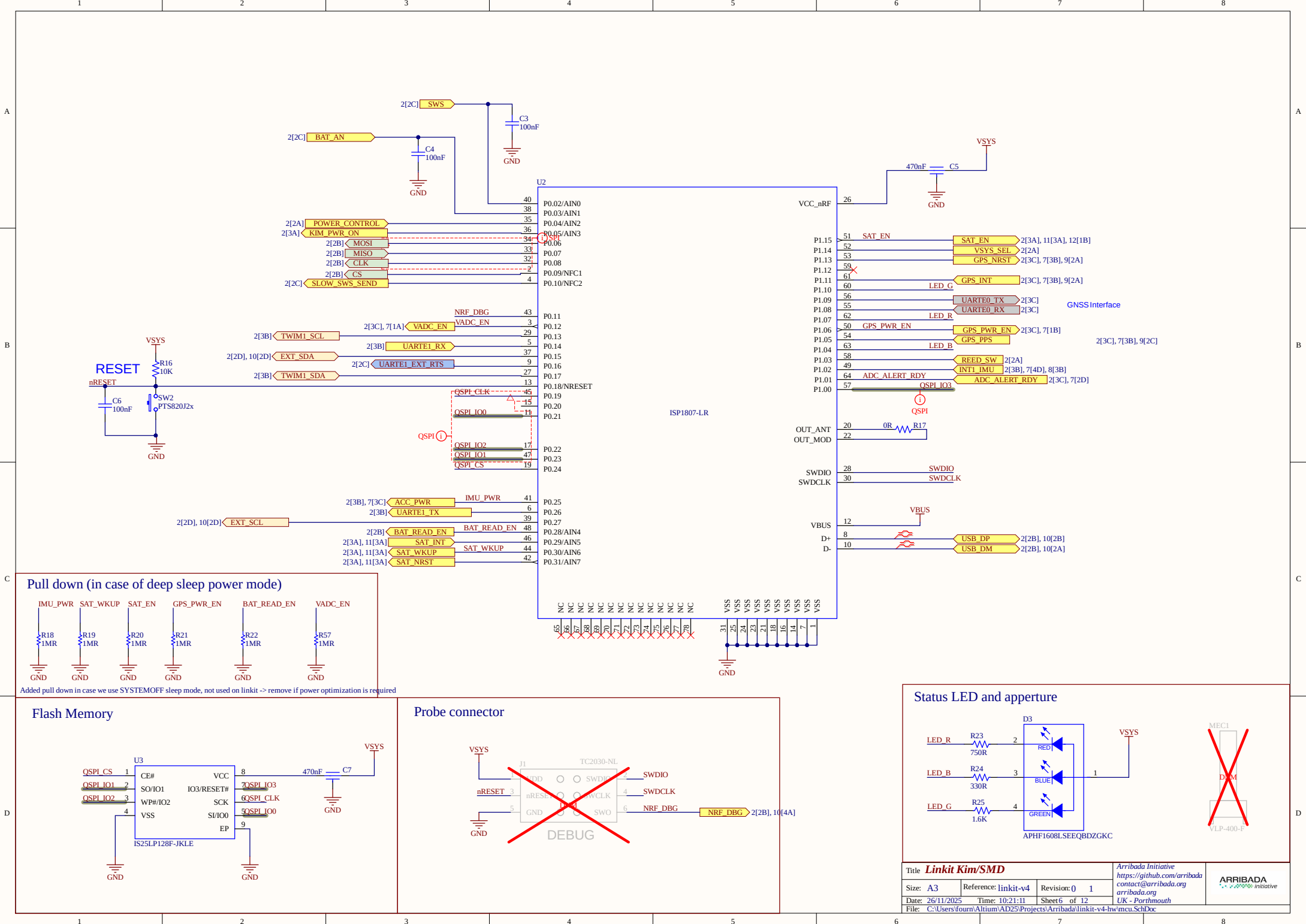


Table 6-3. Output Voltage (SEL Pin = Low) Settings

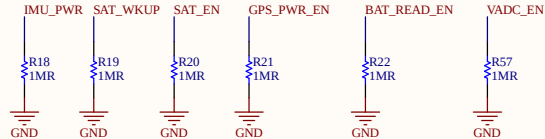
Output Voltage - V _{OUT1} (SEL = LOW)	R _{CFG3}
1.8 V	0 Ω
2.0 V	511 Ω
2.1 V	1.15 kΩ
2.2 V	1.87 kΩ
2.3 V	2.74 kΩ
2.4 V	3.83 kΩ
2.5 V	5.11 kΩ
2.6 V	6.49 kΩ
2.7 V	8.25 kΩ
2.8 V	10.5 kΩ
3.0 V	13.3 kΩ
3.3 V	16.2 kΩ
3.6 V	20.5 kΩ
4.0 V	24.9 kΩ
4.5 V	30.1 kΩ
5.0 V	36.5 kΩ

Output Voltage - V _{OUT1} (SEL = HIGH)	Input Current Limit							
	UNLIMITED	100 mA	50 mA	25 mA	10 mA	5 mA	2.5 mA	1 mA
1.8 V	R _{CFG1}	R _{CFG2}	R _{CFG1}	R _{CFG2}	R _{CFG1}	R _{CFG2}	R _{CFG1}	R _{CFG2}
1.8 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
1.9 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.0 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.1 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.2 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.3 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.4 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.5 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.6 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.7 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.8 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.9 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.0 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.1 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.2 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.3 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.4 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
3.5 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
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4.0 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.1 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.2 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.3 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
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4.5 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.6 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.7 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.8 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
5.0 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ



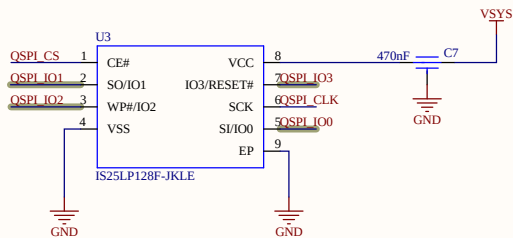


Pull down (in case of deep sleep power mode)

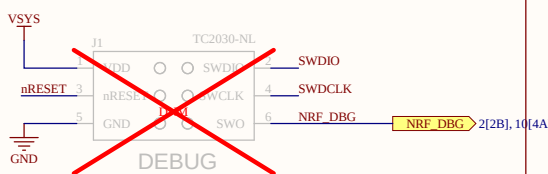


Added pull down in case we use SYSTEMOFF sleep mode, not used on linkit -> remove if power optimization is required

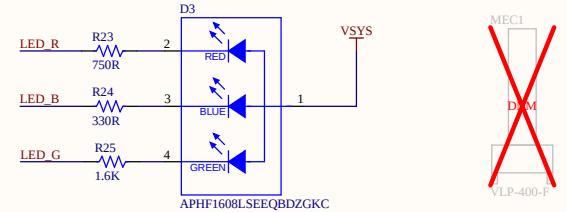
Flash Memory



Probe connector

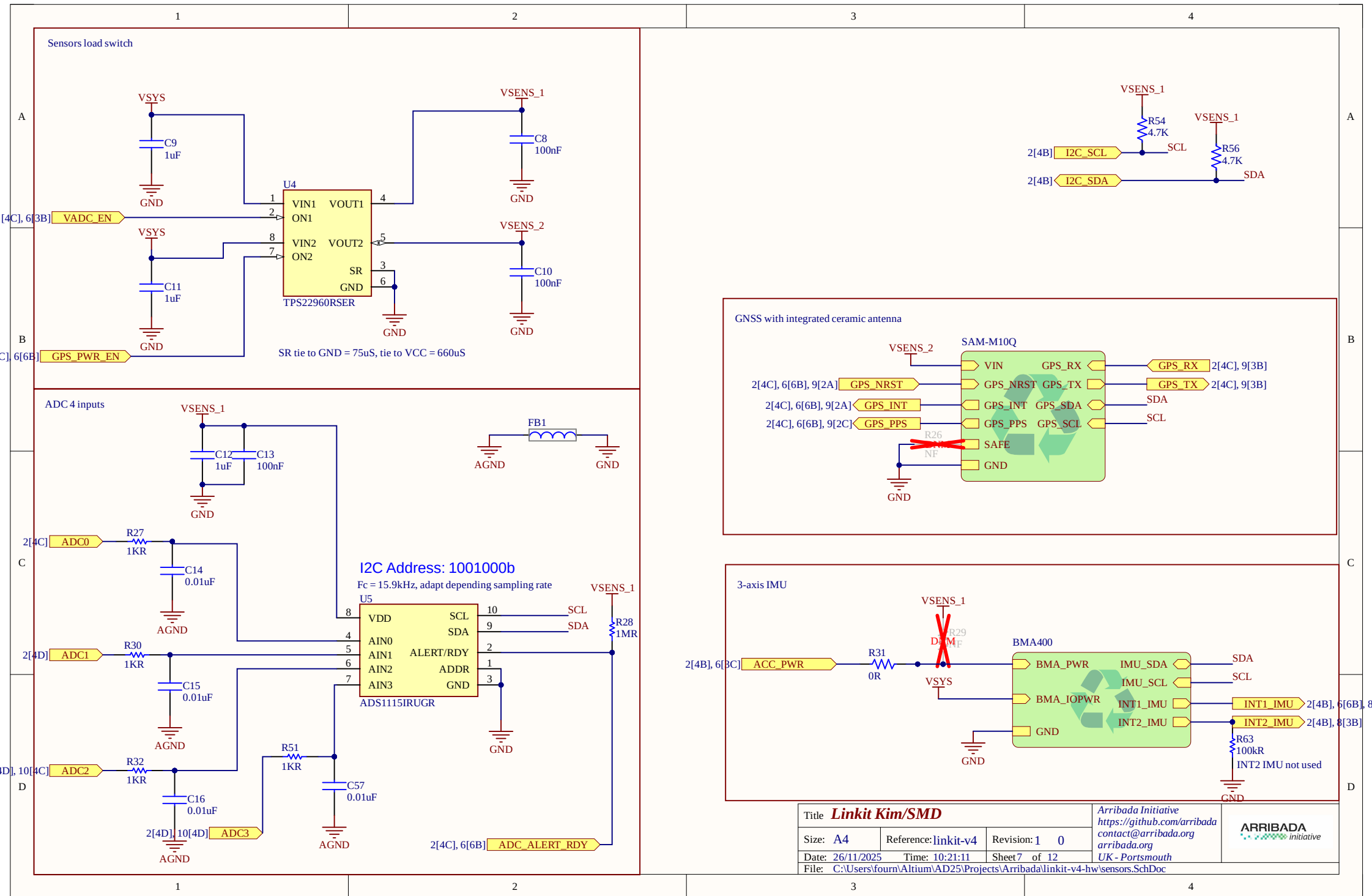


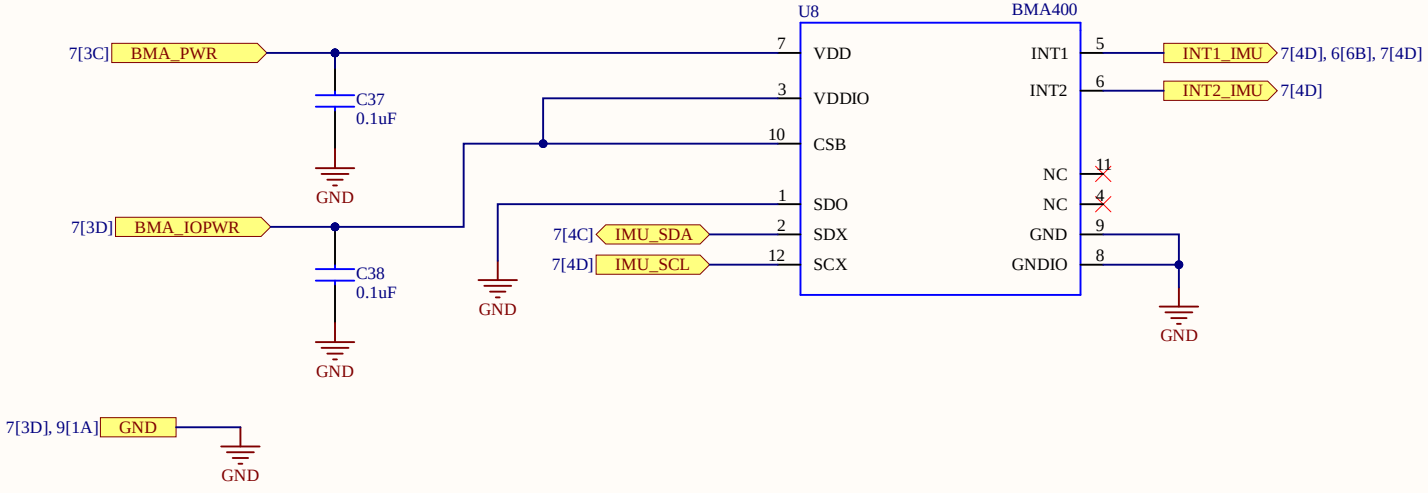
Status LED and aperture

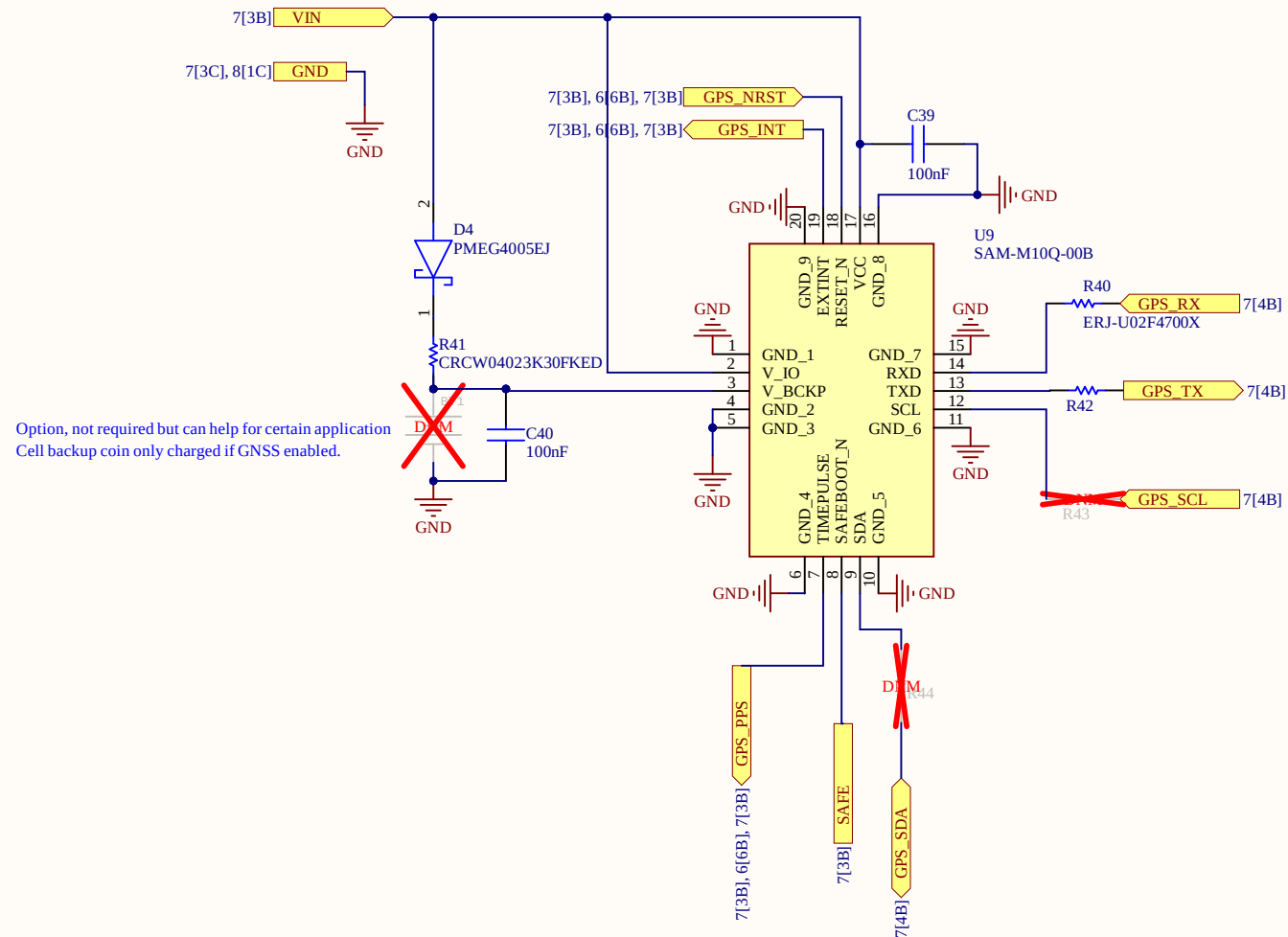


Title Linkit Kim/SMD			Arribada Initiative https://github.com/arribada contact@arribada.org arribada.org		
Size: A3	Reference: linkit-v4	Revision: 0	1		
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File: C:\Users\four\Altium\AD25\Projects\Arribada\linkit-v4-hw\mcu.SchDoc					



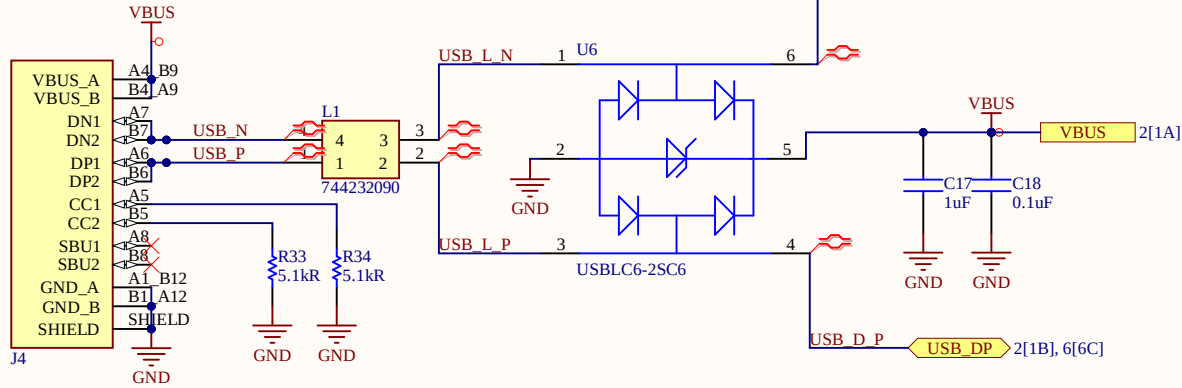




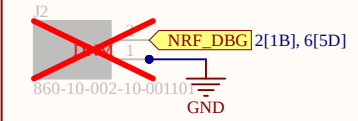


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Size: A4	Reference: linkit-v4	Revision: 1 0	Date: 26/11/2025 Time: 10:21:11 Sheet 9 of 12	
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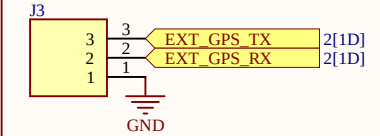
USB Type-C



NRF Debug UART out

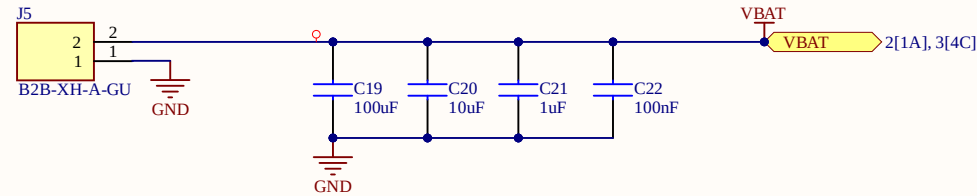


GPS Debug



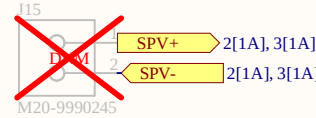
Battery and decoupling

Add polarity indicator

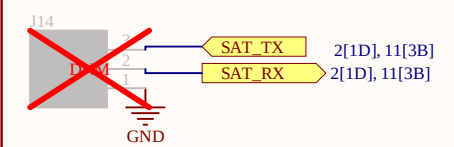


Solar panel input

Add polarity indicator

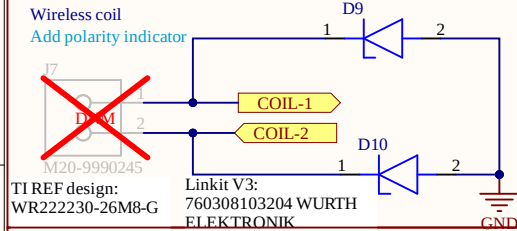


SAT UART



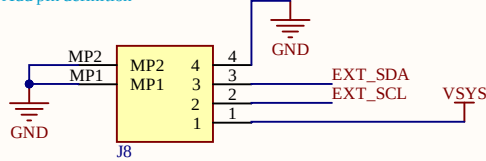
Wireless coil

Add polarity indicator

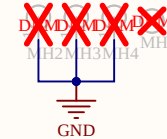


BM04B connector dedicated to bluerobotics thermal sensor

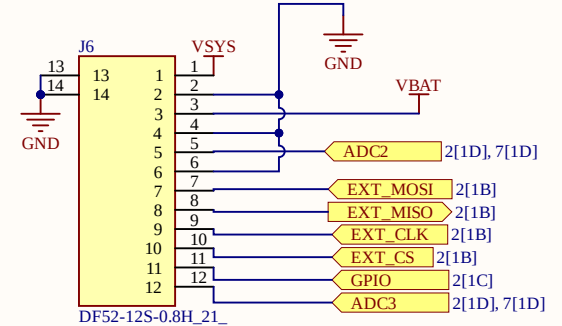
Add pin definition



Fixation VIAs

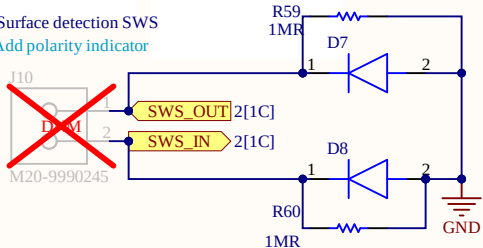


Header interface



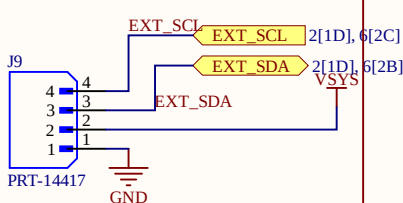
Surface detection SWS

Add polarity indicator



Qwiic I2C connector

Add pin definition



Title **Linkit Kim/SMD**

Size: **A4** Reference: **linkit-v4**

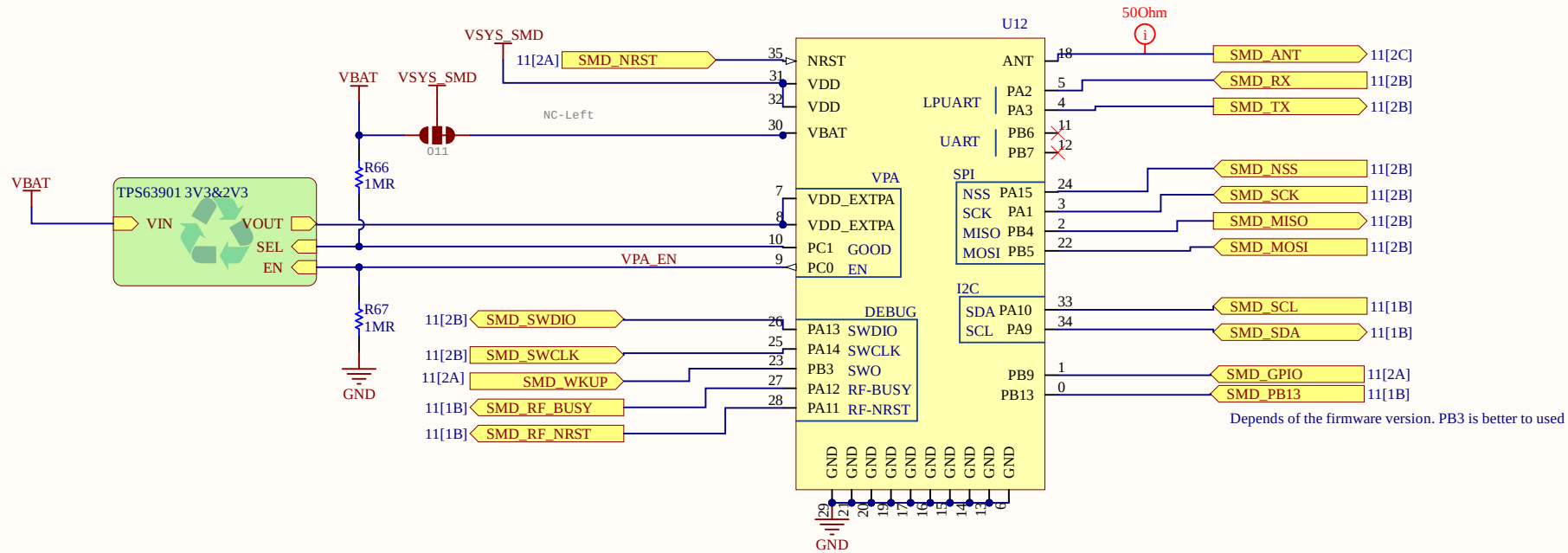
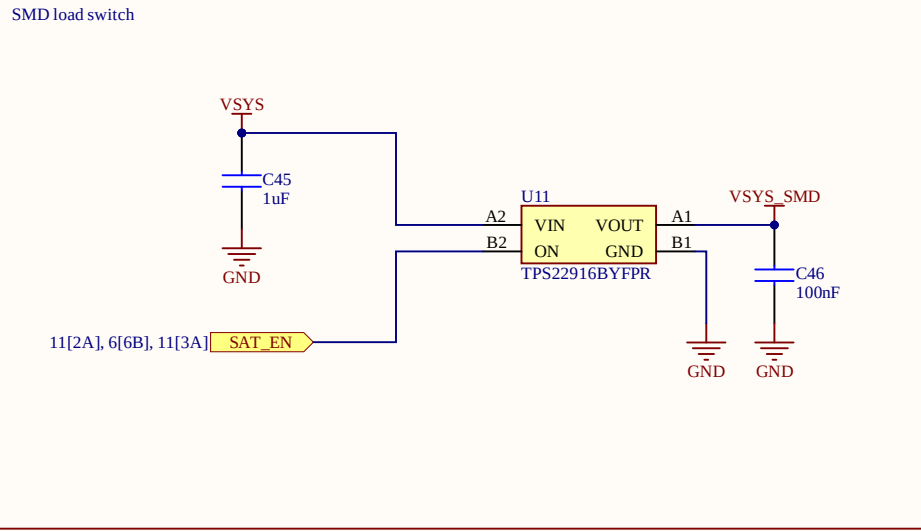
Revision: **1 0**

Date: 26/11/2025 Time: 10:21:12 Sheet 10 of 12

File: C:\Users\fourm\Altium\AD25\Projects\Arribada\linkit-v4-hw\interface.SchDoc

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Size: A4	Reference: linkit-v4	Revision: 1 0		
Date: 26/11/2025	Time: 10:21:12	Sheet 12 of 12		
File: C:\Users\fourn\Altium\AD25\Projects\Arribada\linkit-v4-hw\sat-smd.SchDoc				

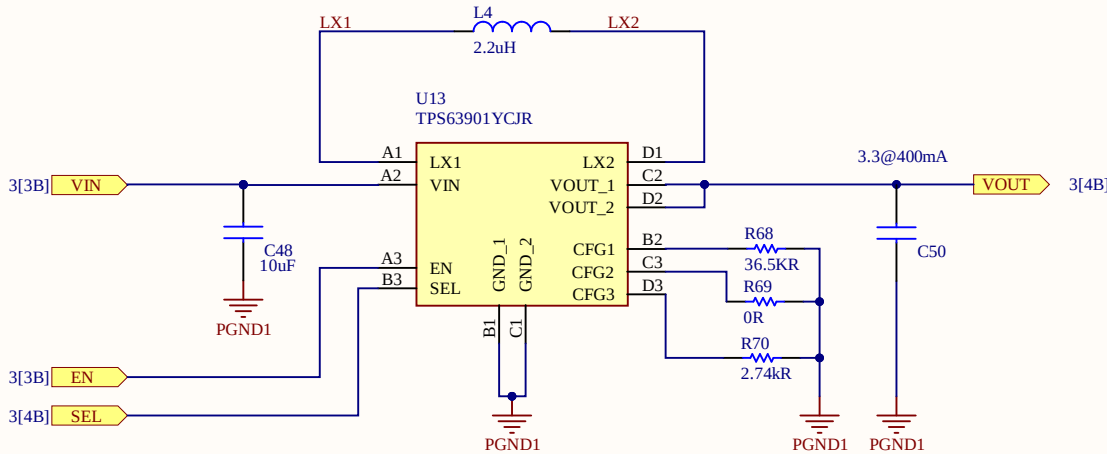
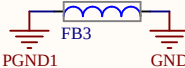


Table 6-3. Output Voltage (SEL Pin = Low) Settings

Output Voltage – V _{OUT1} (SEL = LOW)	R _{CFG3}
1.8 V	0 Ω
2.0 V	511 Ω
2.1 V	1.15 kΩ
2.2 V	1.87 kΩ
2.3 V	2.74 kΩ
2.4 V	3.83 kΩ
2.5 V	5.11 kΩ
2.6 V	6.49 kΩ
2.7 V	8.25 kΩ
2.8 V	10.5 kΩ
3.0 V	13.3 kΩ
3.3 V	16.2 kΩ
3.6 V	20.5 kΩ
4.0 V	24.9 kΩ
4.5 V	30.1 kΩ
5.0 V	36.5 kΩ

Output Voltage – V _{OUT1} (SEL = HIGH)	Input Current Limit							
	UNLIMITED	100 mA	50 mA	25 mA	10 mA	5 mA	2.5 mA	1 mA
1.8 V	R _{CFG1}	R _{CFG2}	R _{CFG1}	R _{CFG2}	R _{CFG1}	R _{CFG2}	R _{CFG1}	R _{CFG2}
1.8 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
1.9 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.0 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.1 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.2 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.3 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.4 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.5 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.6 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
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2.8 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
2.9 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.0 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.1 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.2 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.3 V	0 Ω	511 Ω	1.15 kΩ	1.87 kΩ	2.74 kΩ	3.83 kΩ	5.11 kΩ	6.49 kΩ
3.4 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
3.5 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
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4.2 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.3 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.4 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.5 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.6 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.7 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
4.8 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ
5.0 V	0.25 kΩ	10.5 kΩ	13.3 kΩ	16.2 kΩ	20.5 kΩ	24.9 kΩ	30.1 kΩ	36.5 kΩ



Board Stack Report

Stack Up		Layer Stack			
Layer	Board Layer Stack	Name	Material	Thickness	Constant
1		Top Overlay		0mil	
2		Top Solder	SM-003	1mil	4
3		Top Surface Finish	Nickel, Gold	0.158mil	
4		Top Layer	CF-004	1.378mil	
5		Dielectric 1	S1000-2MB	1.951mil	4.25
6		Dielectric 2	S1000-2MB	1.951mil	4.25
7		Int1 (GND)	CF-004	0.689mil	
8		Dielectric 3	S1000-2MB	1.951mil	4.25
9		Dielectric 4	S1000-2MB	1.951mil	4.25
10		Int2 (Sign)	CF-004	1.378mil	
11		Dielectric 5	Core-001-HCC	41.929mil	4.25
12		Int3 (Sign)	CF-004	1.378mil	
13		Dielectric 6	S1000-2MB	1.951mil	4.25
14		Dielectric 7	S1000-2MB	1.951mil	4.25
15		Int4 (PWR)	CF-004	0.689mil	
16		Dielectric 8	S1000-2MB	1.951mil	4.25
17		Dielectric 9	S1000-2MB	1.951mil	4.25
18		Bottom Layer	CF-004	1.378mil	
19		Bottom Surface Finish	Nickel, Gold	0.158mil	
20		Bottom Solder	SM-003	1mil	4
21		Bottom Overlay		0mil	
	Height : 66.741mil				

